

DONGSHENG (DAVID) HAN, Ph.D.

Postdoctoral Scholar, Stanford Bioengineering • NIH/NIA F99/K00 Fellow

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RESEARCH PROFILE

Biologist-engineer building experimental and computational platforms to measure and reprogram cell state. I combine multiplexed CRISPR perturbation and single-cell genomics with machine learning and agentic AI systems, closing the loop between wet-lab data and computation toward cellular rejuvenation and improved T-cell therapy.

RESEARCH EXPERIENCE

NIH/NIA F99/K00 Postdoctoral Scholar | Stanford University **2025–Present**

- Design epigenetic and metabolic interventions to enhance T-cell function, persistence, and antitumor activity across cancer and aging.
- Engineer multiplexed CRISPR-Cas13d perturbation screens with single-cell and multi-omic readouts to identify regulators of T-cell rejuvenation and exhaustion.
- Develop computational and AI tooling that turns high-dimensional screens into prioritized, testable hypotheses.

Visiting Ph.D. Researcher | University of California, San Francisco **2024–2025**

- Established direct induced-neuron models that preserve aging signatures and optimized CRISPRa/i delivery to mature neurons.
- Applied single-cell transcriptomic and functional profiling to discover candidate neuronal rejuvenation factors.

Ph.D. Researcher | Vanderbilt University **2020–2025**

- Discovered how the chromatin reader BRWD3 promotes proteasomal degradation of KDM5 to maintain H3K4 methylation homeostasis.
- Developed nanopore-based experimental and analytical methods to map active DNA replication at single-molecule resolution.
- Used ChIP-seq, CUT&RUN, ATAC-seq, biochemistry, and cell biology to define BRWD3 functions in chromatin and replication initiation.

Undergraduate & M.S. Researcher | Sun Yat-sen University **2015–2019**

- Defined developmental functions of the RNA m6A methyltransferase METTL16 using CRISPR-engineered Drosophila models.

COMPUTATIONAL & AI PLATFORMS

Agentic target selection

- Built a multi-agent pipeline combining deterministic scoring, automated literature grounding, and orchestration to prioritize gene-combination targets for multiplexed screens.

Perturb-seq analysis

- Developed an end-to-end Python workflow using scanpy, pertpy, pyDESeq2, Mixscape, and scVI/scANVI for weak-knockdown regimes and batch-corrected differential expression.

Machine learning for genomics

- Apply machine learning and gene-regulatory-network inference to single-cell and single-molecule data, including nanopore replication mapping and sequence-design models.

EDUCATION

Ph.D., Molecular Biology — Vanderbilt University, 2025

M.S., Biotechnology Engineering — Sun Yat-sen University, 2019

B.S. with Honors, Biology — Sun Yat-sen University, 2017

SELECTED PUBLICATIONS

1. **Han D**, Schaffner SH, Davies JP, Benton ML, Plate L, Nordman JT. BRWD3 promotes KDM5 degradation to maintain H3K4 methylation levels. Proceedings of the National Academy of Sciences. 2023;120(39):e2305092120. *First author*

2. **Han D**, Shepherd C, Benton ML, Nordman JT. Nanopore-based sequencing of active DNA replication reveals key principles of metazoan replication dynamics. Science Advances. 2026;12(18):eaed2806. *First author*

3. **Han D**, Churcher S, Nordman JT. PCR cloning Intermediated Gibson assembly (PIG) for constructing DNA repair templates in CRISPR-Cas9 based gene editing. microPublication Biology. 2023. *First author*

4. Munden A, Wright MT, **Han D**, Tirgar R, Plate L, Nordman JT. Identification of replication fork-associated proteins in Drosophila embryos and cultured cells using iPOND coupled to quantitative mass spectrometry. Scientific Reports. 2022;12:1–11.

5. Li KK, **Han D**, Chen F, et al. Compensatory replacement of the BigH1 variant histone by canonical H1 supports normal embryonic development in Drosophila. bioRxiv. 2019.

SELECTED HONORS & FUNDING

- **2024–2029** NIH/NIA F99/K00 Fellowship
- **2024** Graduate Research Excellence Award in Biological Sciences, Vanderbilt University
- **2023** Biomedical Research Education & Training Traveling Grant Award, Vanderbilt University
- **2017** Outstanding Poster Award, National Drosophila Research Conference, China
- **2017** Excellent Undergraduate Thesis and B.S. with Honors, Sun Yat-sen University

TECHNICAL EXPERTISE

Functional genomics	CRISPR-Cas9, CRISPR-Cas13d, CRISPRa/i, pooled and arrayed perturbation screens, assay development
Single-cell & genomics	10x scRNA-seq, ChIP-seq, CUT&RUN, ATAC-seq, nanopore sequencing
AI & machine learning	Agentic and multi-agent LLM systems, machine learning for genomics, gene-regulatory-network inference
Programming & analysis	Python, R, Linux, single-cell analysis, differential expression, reproducible pipelines
Cell & molecular biology	Primary cell culture, flow cytometry, confocal microscopy, cloning, qPCR, immunoprecipitation, western blot
Research domains	Epigenetics, DNA replication, aging biology, cellular reprogramming, T-cell engineering

LEADERSHIP & COMMUNICATION

Mentored undergraduate, graduate, and research staff trainees; delivered selected departmental and conference talks; served as a Biological Sciences graduate-student DEI representative and international-researcher volunteer.